

1st REVIEW

FraudShield: Unified Fraud Detection & Risk Scoring Platform

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Name of Candidate	Role
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ABSTRACT

- Digital payments and e-commerce transactions have grown rapidly, increasing the volume and sophistication of financial fraud.
- FraudShield is a Unified Fraud Detection & Risk Scoring Platform combining machine learning with real-time stream processing.
- A Kafka-based streaming pipeline ingests transactions; behavioural features feed an XGBoost classifier tuned for imbalanced fraud data.
- FastAPI, Redis, and PostgreSQL deliver low-latency inference and durable case records.
- SHAP-based explainability and a React dashboard give analysts a clear, real-time view of flagged transactions and cases.

INTRODUCTION

- Financial institutions, payment gateways, and e-commerce platforms process millions of transactions daily.
- Fraud tactics evolve constantly — account takeover, card-not-present fraud, synthetic identities, and velocity-based abuse.
- Machine learning combined with real-time stream processing has become the industry standard for fraud detection.
- This project builds an end-to-end pipeline demonstrating this approach, from data ingestion to analyst-facing dashboard.

EXISTING SYSTEM

- Legacy fraud detection systems rely on static, rule-based thresholds (e.g., flag if amount > fixed limit).
- Rules are manually defined and updated, making them slow to adapt to new fraud patterns.
- High false-positive rates lead to genuine transactions being blocked, frustrating customers.
- No unified system exists for real-time scoring, alerting, and case investigation — teams rely on disconnected tools.
- Limited or no explainability: analysts often cannot see why a rule fired.

PROBLEM

- How can suspicious transactions be identified accurately and in real time, without relying on rigid, static rules?
- How can false positives be reduced while still catching sophisticated, evolving fraud patterns?
- How can fraud analysts be given clear, explainable reasons for every flagged transaction?
- How can monitoring, alerting, and case management be unified into a single platform?

SOLUTION(Contribution)

- FraudShield: a real-time ML-based risk-scoring platform for every transaction.
- A Kafka streaming pipeline ingests transaction events and feeds engineered behavioural features to the model.
- An XGBoost classifier, tuned for class-imbalanced fraud data, produces a 0–100 risk score per transaction.
- FastAPI, Redis, and PostgreSQL together deliver low-latency inference and durable, queryable case records.
- SHAP-based explainability and a React-based Fraud Risk Command Center dashboard give analysts a clear "why flagged" view.

Objectives

- Build a feature-engineered, imbalance-aware ML model that reliably flags fraudulent transactions.
- Design a low-latency pipeline (Kafka → FastAPI → Redis/PostgreSQL) capable of real-time inference.
- Provide a Fraud Risk Command Center dashboard for live monitoring and case triage.
- Make model decisions explainable to non-technical fraud analysts using SHAP.
- Evaluate the system on precision, recall, F1-score, and AUC-ROC.

Modules

- Data Pipeline & ML Model — feature engineering, class-imbalance handling, XGBoost training and tuning, SHAP explanations (D. Rohini Tanmayi).
- Streaming, Backend API & Data Layer — Kafka ingestion, FastAPI endpoints, Redis caching, PostgreSQL schema (S. Manvi Vaishnavi).
- Dashboard & Explainability UI — React-based Fraud Risk Command Center, live monitoring, alerts, and case management views (Varshith).

SOFTWARE REQUIREMENTS

- Programming Languages: Python 3.10+, JavaScript (ES6+)
- ML Libraries: pandas, scikit-learn, XGBoost, SHAP, imbalanced-learn
- Backend: FastAPI, Uvicorn, SQLAlchemy
- Streaming: Apache Kafka 3.x
- Database: PostgreSQL 14+, Redis 7+
- Frontend: React.js, Recharts, Axios
- Tools: VS Code, Git/GitHub, Postman, Docker

CONCLUSION AND FUTURE SCOPE

- FraudShield demonstrates a complete, industry-style fraud detection pipeline from real-time ingestion to explainable, analyst-facing dashboards.
- The system achieves real-time risk scoring, explainable predictions, and unified case management in one platform.
- Future Scope: model retraining pipeline using analyst-confirmed feedback labels.
- Future Scope: graph-based features to detect fraud rings across linked accounts/devices.
- Future Scope: multi-channel alerting (email/SMS/Slack) and cloud-scale deployment with Kubernetes.

THANK YOU